

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning Data intergration, Data transformation and data Reduction for effective data preparation (BL3, BL4)
Assessment Tool Weightage: 20%

QUESTIONS: 10

Total Marks:20

Annexure A

CLASS TEST

Criteria	Understanding of Task (2)	Procedure & Execution (2.5)	Observation & Result (2)	Viva Voce / Conceptual Response (2)	Time Management & Presentation (1.5)	<u>Total(10)</u>
Weightage	20%	25%	20%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						
Q6						
Q7						
Q8						
Q9						
Q10						

Code of Conduct:

I G Sai Deepak(192311432) Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Deepak G

2. What is data cleaning? Mention two common data quality issues. Explain the handle missing values and noisy data with suitable examples.

A) Data Cleaning :

Data cleaning is the process of detecting and correcting (or removing) inaccurate, incomplete, noisy, or inconsistent data from a dataset so that it becomes suitable for reliable analysis and mining.

Two Common data Quality Issues :

- (i) Missing values
- (ii) Noisy data.

Handling missing values :

- ① Ignore the tuple.
- ② Fill in manually.
- ③ Use a global constant
- ④ Use a measure of central tendency.
- ⑤ Use the attribute mean/median of the class.
- ⑥ Use the most probable value.

Handling noisy data :

- ① Binning - Sort the data into bins.
- ② Regression - fit the data to a regression function.
- ③ Clustering : group similar values together.
- ④ Combined Computer and human inspection.

3. Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration.

A. Data Integration:

Data Integration is the process of Combining data from multiple, often heterogeneous, sources into a single, coherent data store such as data warehouse.

Two challenges in data integration:

- i) Entity identification problem.
- ii) Redundancy and Correlation problems.
 - iii) Handling redundancy.
 - a) Correlation analysis.
 - b) Chi-Square Test.

Handling inconsistency:

Inconsistency arises from differences in representation, scaling, encoding, or naming across sources. It is handled by:

- i) Establishing a Common Schema.
- ii) Applying data transformation rules to convert all sources.
- iii) Careful, Symmetric data auditing during integration to detect and resolve value conflicts as sources are combined.

4. What is Covariance Analysis? List the two Covariance formulas.

A) Covariance Analysis:

Covariance Analysis measures how two numerical attributes vary together - i.e., whether an increase in one attribute tends to occur along with an increase (or decrease) in the other. It is used in data integration to detect redundancy between attributes: a Covariance far from Zero indicates the attributes change together and one may be redundant.

Two Covariance formulas:

① Population Covariance: (used when the full dataset of n points is treated as population):

$$\text{Cov}(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n}$$

② Sample Covariance: used when the data is a sample drawn from a larger population - Bessel's Correction.

$$\text{Cov}(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n-1}$$

* A +ve Covariance means x and y tend to increase together.

* A -ve Covariance means x and y tend to increase as the other decreases.

5. Suppose two Stocks A and B have following values in one week: (2,5), (3,8), (5,10), (4,11), (6,14). Question: If the Stocks are affected by same industry needs, will their prices go fall together and independent?

A) Given data ($A=x$, $B=y$)

Day	A(x)	B(y)
1	2	5
2	3	8
3	5	10
4	4	11
5	6	14

$$*(\text{Mean of A}) = \frac{2+3+5+4+6}{5} = 4.0$$

$$*(\text{Mean of B}) = \frac{5+8+10+11+14}{5} = \frac{48}{5} = 9.6$$

Compute deviation and products:

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
2	5	-2	-4.6	9.2
3	8	-1	-1.6	1.6
5	10	1	0.4	0.4
4	11	0	1.4	0.0
6	14	2	4.4	8.8

Step 3 : Covariance:

$$\sum (x - \bar{x})(y - \bar{y}) = 9.2 + 1.6 + 0.4 + 0.0 + 8.8 = 20.0$$

$$\text{Cov}(A, B) = \frac{20.0}{5} = 4.0 \text{ (population Covariance)}$$

6. Find Covariance for the following data set $x = \{4, 5, 6, 7, 9\}$, $y = \{8, 3, 7, 5, 6\}$

A) Step 1: Means

$$\bar{x} = (4 + 5 + 6 + 7 + 9) / 5 = 31 / 5 = 6.2$$

$$\bar{y} = (8 + 3 + 7 + 5 + 6) / 5 = 29 / 5 = 5.8$$

Step 2: Deviations and products

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
4	8	-2.2	2.2	-4.84
5	3	-1.2	-2.8	3.36
6	7	-0.2	1.2	-0.24
7	5	0.8	-0.8	-0.64
9	6	2.8	0.2	0.56

Step 3: Covariance.

$$\sum (x - \bar{x})(y - \bar{y}) = -4.84 + 3.36 - 0.24 - 0.64 + 0.56$$
$$= -1.80$$

$$\text{Cov}(x, y) = -1.80 / 5 = -0.36 \text{ (population covariance)}$$

Since, the Covariance is small and negative, x and y show a very weak inverse (negative) relationship - as x increase slightly, y tends to decrease slightly, though the relationship is weak.

7. Step 1: Sort the data (already sorted) - 27 values.

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

Step 2: partition into equal-frequency bins

<u>Bin</u>	<u>Values</u>	<u>Bin Mean</u>	<u>Bin Boundaries</u> (min, max)
Bin 1	13, 15, 16, 16, 19, 20, 20, 21, 22	18.00	(13, 22)
Bin 2	22, 25, 25, 25, 25, 30, 33, 35	26.11	(22, 35)
Bin 3	35, 35, 35, 36, 40, 45, 46, 52, 70	43.78	(35, 70)

Step 3a: Smoothing by bin means

* Bin 1 $\rightarrow$ all 9 values becomes 18.00.

* Bin 2 $\rightarrow$ all 8 values becomes.

* Bin 3 $\rightarrow$ all 9 values becomes 43.78.

Integration:

Equal-frequency binning places roughly the same no. of records in each bin (here 9 each), unlike equal width binning where the values-range would instead be divided equally. This makes the bins robust to the skew visible in this data (many patients clustering between 20-36, with a sparse high tail up to 70). giving more balanced smoothing than equal-width bins would.

1. Define Data Mining and list any two basic data mining tasks. Explain the steps involved in the KDD process and justify the role of data preprocessing in it.

A) Definition:

Data mining is the process of discovering interesting, non-trivial, previously unknown and potentially useful patterns, correlations, or trends from large volumes of data using statistical, mathematical and machine-learning techniques.

Two basic data mining tasks:

i. Classification.

ii. Clustering.

Steps in KDD:

- ① Data Cleaning.
- ② Data Integration.
- ③ Data Selection.
- ④ Data Transformation.
- ⑤ Data Mining.
- ⑥ Pattern Evaluation.
- ⑦ Knowledge presentation.

Role of data preprocessing in KDD: Data preprocessing corresponds to the first four of KDD and is critical because real-world data is typically incomplete, noisy and inconsistent.